VARUN NARENDRA PATIL

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PROFESSIONAL SUMMARY

M.S. Mechanical Engineering candidate focused on mechanical design, thermal systems, and product development from concept through validated experimentation. Experienced in CAD-led prototyping, thermal-fluid simulation, and bench-scale testing for battery thermal management and multiphase heat-transfer applications. Applies instrumentation, calibration, and optimization workflows to translate design intent into repeatable performance data and manufacturable hardware.

TECHNICAL SKILLS

CAD / Design: SolidWorks, AutoCAD, GD&T, Engineering Drawings, Prototyping
Simulation: ANSYS Fluent, FEA, CFD, Multiphysics Modelling, VOF, Boiling & Condensation
Manufacturing: CNC Machining, 3D Printing, DFM, Lean Manufacturing, Quality Control
Programming / Data / Controls: LabVIEW, MATLAB, Python, Data Analysis, DAQ Integration
Experimental / Instrumentation: Thermocouples, Pressure Sensors, IR Thermography, Flowmeter Temperature Controllers, Calibration, Uncertainty Analysis

EDUCATION

M.S. Mechanical Engineering (Design & Manufacturing) **Aug 2024 - Present**
California State University, Long Beach

B.E. Mechanical Engineering **Jun 2029 - Jun 2023**
Dr. A.P.J. Abdul Kalam University, Indore, India

EXPERIENCE

Graduate Research Assistant, California State University, Long Beach, CA **Jul 2025 - Present**
- Developed a passive EV battery thermal management system that enabled pump-free heat removal and stable operating temperatures by integrating PCM-coupled two-phase thermosyphon loops and validating simulations with test data.
- Led a 5-member team from design to prototype by building and instrumenting benchtop thermal test rigs for repeatable performance evaluation.
- Cut prototype build time by 30% and cost by 10x by redesigning flow-loop test sections in SolidWorks for CNC machining and rapid 3D-printed builds.
- Strengthened experimental reliability by executing transient battery-like discharge tests and enforcing calibration, repeatability, and uncertainty-tracking standards.
- Standardized lab operations and improved safety by creating BOMs, wiring diagrams, and test procedures and training 2 students on LabVIEW, pumps, and DAQ systems.

Volunteer Research Assistant, California State University, Long Beach, CA **Mar 2025 - Jul 2025**
- Assembled and troubleshot three lab instruments with vendor technicians, improving equipment readiness while maintaining safe, reliable operation under changing lab constraints.
- Supported undergraduate researchers by organizing datasets, generating figures, and iterating report content for symposium preparation and technical communication quality.

Trainee Boiler Attendant, Manufacturing Operations, Aryan Paper Mill Pvt. Ltd., Gujarat, India **Sep 2023 - Feb 2024**
- Supported recycling and production operations in a 100+ ton/day packaging facility by following preventive maintenance routines and safety SOPs for stable output.
- Evaluated 10+ ply configurations that increased compression strength from 95 to 180 lbs and contributed to corrugated hospital bed development for shelter deployment.

PROJECTS / RESEARCH

ML Prediction of Flow-Boiling Heat Transfer Coefficient (Micro/Mini Tubes)
- Built and benchmarked RF, GBM, SVM, and DLNN models on 20,727 HTC measurements from 82 studies, achieving MAPE ~5% and R^2 ~0.99.

Passive EV Battery Thermal Management Using PCM + Two-Phase Thermosyphon
- Advanced thesis research on a buoyancy-driven, pump-free loop with nano-enhanced PCM targeting pack temperatures below ~45 C and DeltaT < 5 C.

Electric Scooter Topology Optimization
- Optimized fork leg, handlebars, and deck in Altair Inspire, cutting total mass ~40% while reducing fork leg weight 69% from 3.6 kg to 1.1 kg.

Tensile Strength of Polyurethane Composites for Wearables
- Printed 20+ polyurethane specimens, ran stress-strain testing, and documented formulation-dependent tensile behaviour in Minitab for wearable materials design decisions.

Pool Boiling and Flow Boiling
-Designed and Fabricated Test section along with assembly parts in Al and SS for an experimental loop with handling capacity of 10 bars of water pressure and 0.19 bar of vacuum in the loop also achieving 120 degrees Celsius of heating capacity.